2025 International Conference on Computer, Control, Informatics and its Applications (IC3INA) | 979-8-3315-9188-5/25/$31.00 ©2025 IEEE | DOI: 10.1109/IC3INA68387.2025.11325127

Retrieval-Augmented Large Language Models for a Chronic Kidney Disease Patient Education Chatbot

Elvira Nurfadhilah
Research Center for Data & Information Sciences
National Research and Innovation Agency
Jakarta, Indonesia
ORCID: 0000-0003-2227-331X

Agung Santosa
Research Center for Data & Information Sciences
National Research and Innovation Agency
Jakarta, Indonesia
ORCID: 0000-0002-9827-4397

Prabu Kresna Putra
Research Center for Data & Information Sciences
National Research and Innovation Agency
Jakarta, Indonesia
ORCID: 0000-0001-9246-8863

***Abstract*—Chronic Kidney Disease (CKD) is rising globally, underscoring the need for scalable and accurate patient education. This paper presents the design and evaluation of a CKD education chatbot specifically for hemodialysis cases in Indonesia that runs on-premises and employs a Retrieval-Augmented Generation (RAG) framework. We conducted a comparative study of a general-purpose LLM (Llama 3.1) and a language adapted LLM (Sahabat AI). Both models were used in identical RAG pipelines and paired with several retrieval embeddings. Using 70 unique questions evaluated across 10 runs per configuration (n = 700), we assessed semantic similarity (METEOR), factual correctness, and RAG-specific metrics (faithfulness, answer relevancy, context precision) with a Gemini 2.0 Flash evaluator. RAG consistently improved performance over non-RAG baselines, with statistically significant gains across metrics. Sahabat AI outperformed Llama 3.1, posting the highest METEOR (0.4116, `mContriever`) and correctness (4.0293 with `BGE-M3`). It also exhibits higher, more stable faithfulness (∼0.95), indicating closer adherence to the source evidence. These findings suggest that, for healthcare education, localized LLMs combined with carefully chosen embeddings can yield more effective and trustworthy systems.**

***Index Terms*—Chatbot; Chronic Kidney Disease; Hemodialysis; Large Language Models; Retrieval-Augmented Generation.**

I. INTRODUCTION

Chronic Kidney Disease (CKD) is an escalating global health problem, particularly at advanced stages where hemodialysis is often required as a life-saving intervention. In 2024, an estimated 134,057 individuals underwent hemodialysis in Indonesia, underscoring the national scale of the challenge [1]. The economic burden is substantial; Indonesia's National Health Insurance (BPJS Kesehatan) reportedly allocated approximately IDR 11 trillion in 2024 for CKD-related treatment [1]. Major risk factors include hypertension, diabetes, obesity, and population aging, jointly driving increases in CKD incidence and prevalence.

Timely detection and patient education are crucial to slowing disease progression and reducing healthcare costs. Conventional education approaches often suffer from limited reach and barriers to understanding medical terminology. In this context, artificial intelligence (AI)—particularly chatbots—offers transformational potential; however, challenges remain regarding medical accuracy, prevention of model hallucinations, and alignment with up-to-date clinical guidelines, all of which must be addressed systematically [2].

To tackle these challenges, this study introduces *Chatbot Lisa*, an AI solution that evolved from a rule-based system (version 1.0) to a system based on Large Language Models (LLMs) combined with Retrieval-Augmented Generation (RAG) (version 2.0) [3], [4]. RAG was selected for its ability to anchor LLM outputs to verified external knowledge sources, thereby reducing the risk of inaccuracies [5]. Integrating LLM+RAG enables the delivery of medical information that is contextual, current, and source-traceable, thereby enhancing patient engagement and understanding, including support for dietary adherence and lifestyle modifications essential in CKD.

CKD management is resource-intensive, especially at end-stage, which requires ongoing treatments such as hemodialysis [6]. Many patients remain undiagnosed until advanced stages, highlighting the urgency of early detection and sustained education [7]. Specifically, RAG addresses the limitations of generic LLMs by injecting relevant external evidence during inference, improving response accuracy and alignment with the latest clinical guidelines [8]. In nephrology—where practice heavily depends on clinical literature and up-to-date guidelines—the combination of LLMs and RAG can enrich real-time patient education on complex topics such as dialysis management and dietary interventions [7], [8].

Despite rapid adoption of LLMs and RAG in medical domains, important gaps persist: (i) comparative evaluations between global LLMs and linguistically–culturally localized LLMs within a specific medical domain; and (ii) deployment and validation of advanced RAG architectures in healthcare contexts within developing countries such as Indonesia. The contributions of this paper are: (1) presenting the implementation of a hemodialysis-focused medical chatbot built on a localized LLM (Sahabat AI); (2) providing an empirical comparative analysis of a global LLM (Llama 3.1) vs. a local LLM (Sahabat AI) within a RAG framework; and (3) delivering a quantitative evaluation that demonstrates the system's viability as a CKD patient education tool in Indonesia.

Authorized licensed use limited to: b-on: Instituto Politecnico do Porto. Downloaded on November 02,2026 at 15:47:37 UTC from IEEE Xplore. Restrictions apply.

II. METHODOLOGY

The methodology was designed to systematically build and evaluate a hemodialysis-focused medical chatbot, encompassing system-architecture selection, experimental design, and a quantitative evaluation framework across key performance dimensions.

A. LLM Generator

The generator component was responsible for producing responses. Two LLMs served as the foundation models:

1) **Llama 3.1.** A foundation LLM from Meta with strong benchmark performance and a long-context window suitable for complex medical information [9]. It was a general-purpose model with solid reasoning and contextual understanding, though it might not have fully captured Indonesian local nuances.
2) **Sahabat AI.** A Llama-architecture model that underwent continued pre-training on Indonesian and local-language corpora, which made it more sensitive to linguistic–cultural context and local entities [10].

The selection of these LLMs was motivated by three factors: **(1) technical capability**—as a cutting-edge global LLM, Llama 3.1 offered strong reasoning and a long-context window, both relevant for processing complex medical content [9]; **(2) local relevance**—Sahabat AI's continued pre-training on Indonesian and local corpora yielded more accurate clinical-terminology handling and a more natural style for local users [10]; and **(3) data security & sovereignty**—open-source, on-premises deployment ensured full control over data storage and access, reducing leakage or unauthorized-access risks in healthcare settings [11].

B. Retrieval-Augmented Generation (RAG) Architecture

RAG improved LLM accuracy and factuality by integrating external knowledge at inference time, turning a "closed-book exam" into an "open-book exam" [5], [12], [13]. The main flow comprised:

1) **Retrieval.** The system searched for the most relevant context from an external knowledge base using a retriever based on embedding models.
2) **Augmented Generation.** Selected context snippets were injected into the prompt to constrain the LLM's answer space to validated evidence, reducing hallucinations and improving source traceability [5], [12], [13].

C. Embedding Models

Embedding models converted text into vectors so that semantic similarity could be computed during retrieval.

- **`BGE-M3`**. A multilingual, open-source embedding with multi-objective training (multi-purpose, multi-granularity) that was effective for semantic search, including Indonesian.
- **`mContriever`.** A multilingual, label-free contrastive embedding *for unsupervised dense retrieval* that was effective at distinguishing fine-grained meaning for specific queries [14].

TABLE I
STATISTICS FOR THE QA DATASETS.

Dataset / Source	Count
Halodoc (articles)	456
Alodokter (articles)	2,943
Alodokter (QA pairs)	5,886
KPCDI (articles)	204
Knowledge-base PDFs (papers, regs, guidelines)	various
Doctor-validated Q&A (8 categories)	70

- **`ft-BGE-M3`.** A domain-adapted fine-tuning of `BGE-M3` on hemodialysis corpora that yielded sentence embeddings which better captured domain terminology and semantic relations, improving retrieval precision.

III. EXPERIMENT

A. Dataset

We compiled dataset resources across three types: (1) open-domain QA (Alodokter QA pairs and doctor-validated Q&A); (2) articles from trusted health sites (Halodoc, Alodokter, KPCDI); and (3) knowledge-base PDFs covering scientific publications, government regulations (Ministry of Health), and nephrology consortium guidelines. **Table I** reports statistics for the corpora used.

B. RAG

The experiment was designed to evaluate a Retrieval-Augmented Generation (RAG) pipeline using LlamaIndex with Chroma as the vector database. All documents were preprocessed through semantic chunking, in which the corpus was segmented into semantically coherent units (e.g., paragraphs or sections) rather than fixed token windows, ensuring that each chunk preserved natural meaning boundaries. These semantic chunks were embedded using multiple embedding models and stored in Chroma. The retriever was configured with semantic similarity search and a top-k value of 5, so that for each query, the five most relevant semantic chunks were consistently returned.

The workflow proceeded as follows: documents were semantically chunked, embedded, and indexed; queries were executed to retrieve the top-5 chunks; answers were generated based on the retrieved context; and both retrieval and generative quality were evaluated against a gold-standard dataset. The entire pipeline was executed on an NVIDIA A100 GPU, which provided the computational power needed for efficient large-scale embedding, indexing, and evaluation.

C. Evaluation Framework

Experimental design. (i) *Scenario 1—Effect of RAG*: compared base LLM vs. LLM+RAG; metrics were METEOR and Correctness (LlamaIndex) [12], [15]. (ii) *Scenario 2—Embedding Performance*: compared multiple embedding models within RAG; metrics were Correctness, Faithfulness, Answer Relevancy, and Context Precision (Ragas) [13].

Evaluation metrics. We assess both generation quality and retrieval quality using five metrics:

(1) METEOR [15] measures linguistic quality and adequacy of the model response against a reference. Unigram

Authorized licensed use limited to: b-on: Instituto Politecnico do Porto. Downloaded on November 02,2026 at 15:47:37 UTC from IEEE Xplore. Restrictions apply.

TABLE II
QUANTITATIVE EVALUATION RESULTS BY LLM BASE AND EMBEDDING (mean ± std).

LLM Base	Embedding	METEOR	Correctness	Faithfulness	Answer Relevancy	Context Precision
Llama 3.1	—	0.2157 ± 0.0867	3.1717 ± 1.0867	–	–	–
Sahabat AI	—	0.2922 ± 0.0977	3.9257 ± 0.8637	–	–	–
Llama 3.1	BGE-M3	0.3228 ± 0.1826	3.6779 ± 1.0728	0.8796 ± 0.2113	0.6531 ± 0.1286	0.3654 ± 0.4070
Sahabat AI	BGE-M3	0.3921 ± 0.2680	4.0293 ± 0.9469	0.9397 ± 0.1578	0.6588 ± 0.1277	0.3907 ± 0.3524
Llama 3.1	mContriever	0.2903 ± 0.1842	3.6471 ± 1.0766	0.8240 ± 0.2494	0.6586 ± 0.1258	0.2964 ± 0.4247
Sahabat AI	mContriever	0.4116 ± 0.2740	3.9721 ± 1.1324	0.9452 ± 0.1617	0.6535 ± 0.1345	0.3570 ± 0.3591
Llama 3.1	ft-BGE-M3	0.2749 ± 0.1738	3.5429 ± 1.0992	0.8043 ± 0.2779	0.6527 ± 0.1418	0.3231 ± 0.4578
Sahabat AI	ft-BGE-M3	0.3736 ± 0.2644	3.8843 ± 1.1337	0.9446 ± 0.1429	0.6466 ± 0.1280	0.4150 ± 0.3519

matches (exact/stem/synonym/paraphrase) are aligned to compute precision P and recall R, combined via a weighted F_{mean} and penalized by fragmentation:

$$F_{\text{mean}} = \frac{P \cdot R}{\alpha P + (1-\alpha)R}, \text{METEOR} = F_{\text{mean}}\big(1-\gamma(ch/m)^{\beta}\big),$$

where ch is the number of matching chunks and m the number of matched unigrams; α, β, γ follow [15].

(2) Correctness quantifies factual accuracy with respect to the gold answer. For Scenario 1 we use *LlamaIndex CorrectnessEvaluator* [12] which provides an LLM-graded score on a **1–5** scale (higher is better) and a pass/fail flag given a threshold (we use τ=4). For Scenario 2 we use Ragas correctness/factuality scoring [13].

(3) Faithfulness / Groundedness (Ragas) [13] evaluates whether an answer's key atomic claims are *entailed* by the retrieved context; scores are in $[0, 1]$ (higher is better). The pipeline decomposes the answer into claims, maps each claim to candidate evidence from the retrieved context, verifies entailment with an NLI/LLM-based judge, then aggregates outcomes. In node-based RAG, each context chunk is a *node*; a claim is *grounded* if at least one linked node entails it, otherwise it is a hallucination. Thus, the reported score directly reflects the answer's groundedness with respect to the retrieved nodes.

(4) Answer Relevancy (Ragas) [13] quantifies how well a system's response fulfills the question's intent by applying an LLM-based rubric over three aspects—on-topicness (focus on the query), sufficiency/coverage (completeness of the required information), and conciseness/non-rambling (absence of irrelevant or verbose content). The judge produces per-dimension scores in $[0, 1]$, and the final relevancy score is obtained by averaging these values (higher indicates better alignment). This metric is context-agnostic (i.e., independent of retrieved evidence) and can be extended in multi-turn scenarios to include instruction adherence and consistency with prior turns.

(5) Context Precision (Ragas) [13] captures retrieval "cleanliness," i.e., the proportion of retrieved chunks that are relevant to the question/ground truth:

$$\text{Context Precision} = \frac{|C_{\text{rel}}|}{|C|},$$

with C the retrieved set and C_{rel} the subset judged relevant.

IV. RESULTS & DISCUSSION

A. Results

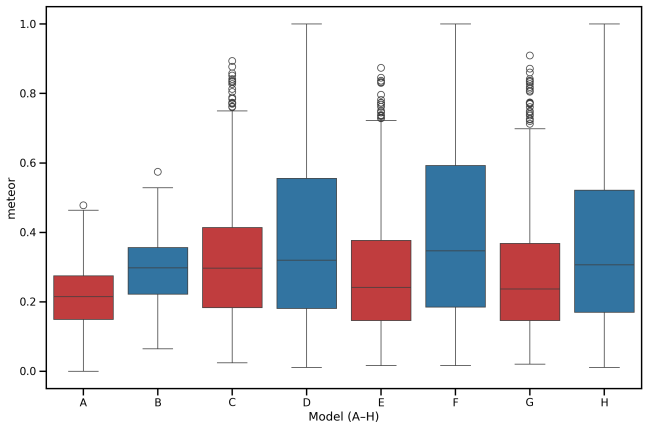
This subsection presented comparative experimental results for two LLMs (Llama vs. Sahabat AI) at baseline and after RAG integration, as well as the effects of embedding models (BGE-M3, mContriever, ft-BGE-M3) on core and RAG-specific metrics. The numerical summary for all configurations was shown in **Table II**, while metric distributions were given in **Figure 1**. The data design used 70 questions × 10 repetitions per configuration (n=700). Evaluation was conducted with Gemini 2.0 Flash as the LLM backend.

At the baseline without RAG, Sahabat AI outperformed Llama 3.1 on two core metrics. Sahabat AI achieved a METEOR of 0.2922 vs. Llama's 0.2157, while Correctness was 3.9257 vs. Llama's 3.1717. Two-sample t-tests on metrics yielded $p \approx 0$ (machine precision), which indicated significant differences and confirmed that the base LLM quality was the determinant of QA performance prior to RAG integration.

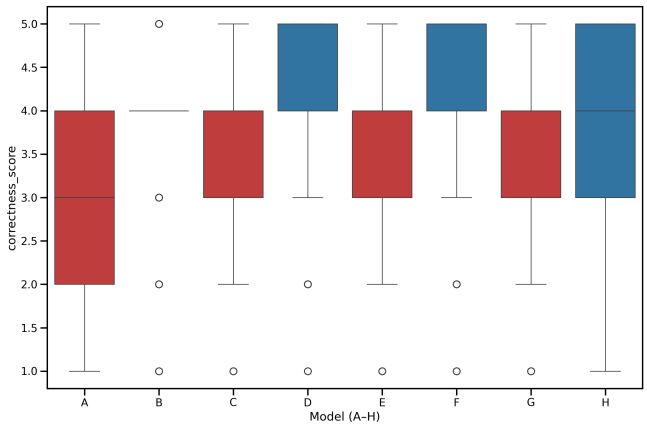
Once RAG was enabled, universal improvements over the baseline were observed ($p \approx 0$ across comparisons). For METEOR, Llama's best configuration was llama-BGE-M3 (0.3228) from a baseline of 0.2157; Sahabat AI peaked with mContriever (0.4116) from a baseline of 0.2922. For Correctness, Llama rose to 3.6779 with BGE-M3 (its highest), and Sahabat AI reached 4.0293 also with BGE-M3 (its highest), which underscored RAG's role in boosting factual accuracy.

Embedding comparisons within RAG revealed model-specific patterns. For Llama, the METEOR ordering was BGE-M3 > mContriever > ft-BGE-M3, with BGE-M3 vs. mContriever differences being significant ($p = 9.53 \times 10^{-4}$). For Sahabat AI, mContriever (METEOR 0.4116; Correctness 3.9721) vs. BGE-M3 (0.3921; 4.0293) was not significantly different (METEOR $p = 0.1783$; Correctness $p = 0.3063$), while ft-BGE-M3 was consistently the lowest on both LLMs. RAG-specific metrics reinforced these findings: Sahabat AI's *Faithfulness* was high and stable ($\approx$ 0.9397–0.9452) compared with Llama ($\approx$ 0.8043–0.8796), with between-LLM tests yielding $p \approx 0$; *Answer Relevancy* tended to plateau around $\approx$ 0.6466–0.6588, with many comparisons not significant; and *Context Precision* showed an LLM×embedding interaction—Sahabat AI+mContriever was superior (mean $\approx$ 0.3570), with, for example, Sahabat AI-BGE-M3 vs. llama-mContriever being significant ($p = 6.83 \times 10^{-6}$).

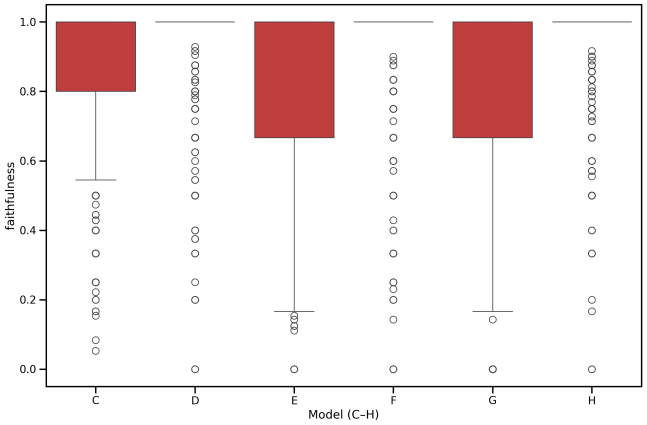
Authorized licensed use limited to: b-on: Instituto Politecnico do Porto. Downloaded on November 02,2026 at 15:47:37 UTC from IEEE Xplore. Restrictions apply.



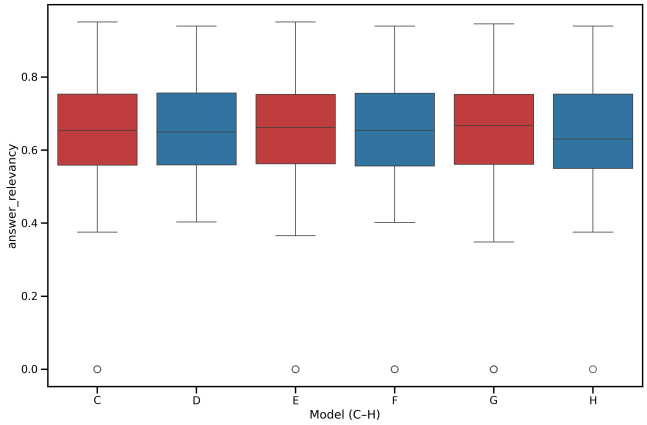
(a) **METEOR**



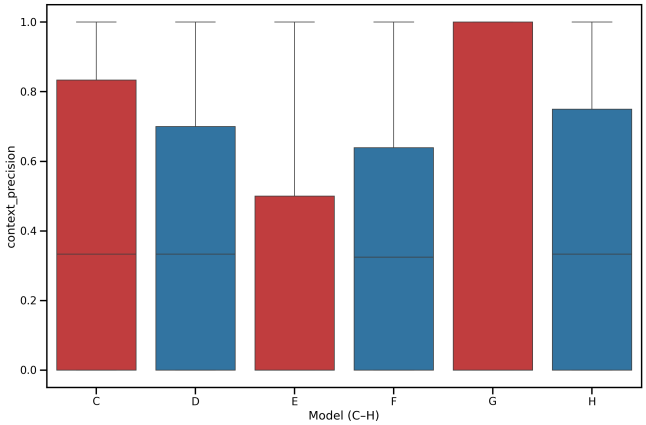
(b) **Correctness**



(c) **Faithfulness**



(d) **Answer Relevancy**



(e) **Context Precision**

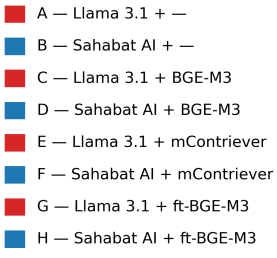


Fig. 1. Metric distributions for all configurations: (a) METEOR, (b) Correctness, (c) Faithfulness, (d) Answer Relevancy, (e) Context Precision.

Authorized licensed use limited to: Inst. Teknologi Sepuluh Nopember. Downloaded on November 02,2026 at 15:47:37 UTC from IEEE Xplore. Restrictions apply.

In summary, Sahabat AI was consistently stronger, RAG helped across the board (especially for Correctness), and `BGE-M3` and `mContriever` were the most reliable embeddings, whereas `ft-BGE-M3` was best avoided in this setting.

B. Discussion

Grounded in the quantitative findings, the system's performance exhibited a clear hierarchy of determinants. The base LLM quality emerged as the dominant factor that set the performance "ceiling"; at the no-RAG baseline, Sahabat AI consistently surpassed Llama 3.1 on METEOR and Correctness, with statistical tests indicating significant differences ($p \approx 0$). This underscored that selecting a linguistically–culturally localized base model established a realistic upper bound for medical question–answering systems.

On this foundation, enabling RAG acted as a second-stage lever that universally improved fluency and factuality (baseline vs. RAG: $p \approx 0$), which indicated that grounding LLM outputs in validated external evidence effectively suppressed hallucinations and strengthened answer traceability. Mechanistically, the consistent rise in Correctness aligned with Sahabat AI's higher and more stable *Faithfulness* relative to Llama, which suggested more effective use of retrieved context when evidence was injected into prompts.

Not all metrics were equally sensitive to architectural variations: *Answer Relevancy* tended to plateau around ~ 0.65, with many across-configuration comparisons not significant, which implied that question-intent design and prompting schemes might have dominated over LLM or embedding choice for this metric. By contrast, *Context Precision* indicated an LLM$\times$embedding interaction; the Sahabat AI+`mContriever` combination yielded higher context precision, which suggested synergy between embedding characteristics and the model's approach to formulating and exploiting retrieval signals. Meanwhile, the domain-tuned `ft-BGE-M3` underperformed in this setup, ranking lowest on *Faithfulness* and *Context Precision*, especially when paired with Llama.

V. CONCLUSION

This study established a clear performance determinant hierarchy: the base LLM set the ceiling, RAG was the primary lever for improving factuality and traceability, and embedding choice tuned retrieval specificity. In Indonesian-language CKD education, the localized LLM (Sahabat AI) consistently surpassed the global model (Llama 3.1) on METEOR and Correctness. Enabling RAG yielded universal gains over the baseline, which indicated the effectiveness of grounding in external evidence to reduce hallucinations. At the component level, `BGE-M3` and `mContriever` were the most stable embedding options, while `ft-BGE-M3` tended to be inferior under the tested setup. Nevertheless, *Answer Relevancy* exhibited a plateau (~ 0.65), which suggested that interventions on intent engineering, query rewriting, and explicit instructions regarding assumptions/constraints/relevance criteria might have been more impactful than simply switching embeddings.

For implementation in Indonesian healthcare, we recommended adopting a localized model as the foundation and treating RAG as the default configuration. Practitioners chose `BGE-M3` or `mContriever` while considering cost–latency trade-offs, and enforced disciplined knowledge-base governance (curation, source dating, periodic updates) so that grounding stayed aligned with the latest clinical guidelines. Teams operationalized human-in-the-loop mechanisms for ambiguous cases, added safety guardrails (clinical disclaimers, risky-intent detection, escalation to clinicians), and monitored quality metrics (Correctness/Faithfulness) alongside operational metrics (latency, per-query cost). To move beyond the relevancy plateau, efforts focused on clarifying intent, automatic query rewriting, and explicit instructions regarding assumptions, limits, and user-expected relevance criteria.

Priority future directions will include broadening scope beyond hemodialysis to the broader CKD spectrum with multisite evaluations across diverse populations; integrating clinician assessments and prospective studies linking chatbot use to behavior change (diet/medication adherence) and clinical outcomes; exploring hybrid RAG (distinct embeddings for initial retrieval vs. reranking) together with intent-understanding modules to further improve relevance; and strengthening reliability via safe personalization (on-premises/federated), uncertainty estimation, and out-of-distribution detection. Technically, model distillation/quantization for constrained devices, multimodal integration (e.g., lab charts, visual education materials), and scheduled knowledge updates will warrant investigation to keep the system efficient, safe, and aligned with evolving guidelines.

ACKNOWLEDGMENT

We express our gratitude to colleagues at the Research Center for Data and Information Sciences, BRIN—Ms. Lyla Ruslana Aini, Mr. M. Teduh Uliniansyah, Ms. Radhiyatul Fajri, and Ms. Siska Pebiana—for their assistance with experiments and data processing. We also thank researchers at the Research Center for Preclinical and Clinical Medicine, BRIN, for supplying reliable data references and validating the test datasets.

REFERENCES

[1] CNN Indonesia, "Pasien cuci darah di Indonesia meningkat, capai 134 ribu di 2024," Mar. 13, 2025. [Online]. Available: https://www.cnnindonesia.com/gaya-hidup/20250313093144-255-1208269/pasien-cuci-darah-di-indonesia-meningkat-capai-134-ribu-di-2024 (accessed Aug. 23, 2025).

[2] A. J. Thirunavukarasu, D. S. J. Ting, K. Elangovan, L. Gutierrez, T. F. Tan, and D. S. W. Ting, "Large language models in medicine," *Nature Medicine*, vol. 29, no. 8, pp. 1930–1940, 2023, doi: 10.1038/s41591-023-02448-8.

[3] Badan Riset dan Inovasi Nasional (BRIN), "Chatbot Lisa (Hemodialisa) versi 1.0," Hak Cipta No. EC00202490266, Direktorat Jenderal Kekayaan Intelektual, Kementerian Hukum dan HAM RI, 2024.

[4] Badan Riset dan Inovasi Nasional (BRIN), "Chatbot Lisa (Hemodialisa) versi 2.0 berbasis LLM+RAG," Hak Cipta No. EC002025048065, Direktorat Jenderal Kekayaan Intelektual, Kementerian Hukum dan HAM RI, 2025.

[5] P. Lewis, E. Perez, A. Piktus, *et al.*, "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 33, pp. 9459–9474, 2020.

Authorized licensed use limited to: b-on: Instituto Politecnico do Porto. Downloaded on November 02,2026 at 15:47:37 UTC from IEEE Xplore. Restrictions apply.

[6] L. Sunariyanti, *et al.*, "Quality of Life of Patients with Chronic Kidney Disease Undergoing Hemodialysis," *Journal of Public Health Research*, vol. 12, no. 3, 2023, doi: 10.1177/22799036231182296.
[7] P. C. Acharya, *et al.*, "AI-driven patient education in chronic kidney disease: Evaluating chatbot responses against clinical guidelines," *Diseases*, vol. 12, no. 8, p. 185, 2024, doi: 10.3390/diseases12080185.
[8] J. Miao, *et al.*, "Integrating Retrieval-Augmented Generation with Large Language Models in Nephrology: Advancing Practical Applications," *Nephrology Journal*, vol. 58, no. 3, pp. 214–225, 2024, doi: 10.3390/npj580324.
[9] Meta AI, "Introducing Meta Llama 3.1," 2024. [Online]. Available: https://ai.meta.com/blog/meta-llama-3-1/
[10] Sahabat AI and GoToCompany, "Model Card for Sahabat-AI/llama3-8b-cpt-sahabatai-v1-base," Hugging Face, 2024. [Online]. Available: https://huggingface.co/Sahabat-AI/llama3-8b-cpt-sahabatai-v1-base
[11] Xite.AI, "Open-Source LLM On-Premise: Ensuring Data Privacy In The Age Of AI," 2024. [Online]. Available: https://xite.ai/blogs/open-source-llm-on-premise-ensuring-data-privacy-in-the-age-of-ai/
[12] J. Liu, "LlamaIndex: A Data Framework for LLM Applications," GitHub repository, 2024. [Online]. Available: https://github.com/run-llama/llama_index
[13] S. Es, J. Shah, D. Mishra, *et al.*, "Ragas: A Framework for Reference-Free Evaluation of RAG Systems," 2023, arXiv:2309.15217.
[14] G. Izacard, M. Caron, L. Hosseini, S. Riedel, P. Bojanowski, A. Joulin, and E. Grave, "Unsupervised Dense Information Retrieval with Contrastive Learning," 2021, arXiv:2112.09118, doi: 10.48550/ARXIV.2112.09118. [Online]. Available: https://arxiv.org/abs/2112.09118
[15] S. Banerjee and A. Lavie, "METEOR: An Automatic Metric for MT Evaluation with Improved Correlation with Human Judgments," in *Proc. ACL Workshop on Intrinsic and Extrinsic Evaluation Measures for Machine Translation and/or Summarization*, 2005.

Authorized licensed use limited to: b-on: Instituto Politecnico do Porto. Downloaded on November 02,2026 at 15:47:37 UTC from IEEE Xplore. Restrictions apply.